Accelerometer based Complex Nurse Care Activity Recognition using Machine Learning Approach

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ABSTRACT

Researchers have been working for a long time to recognize human activities based on sensor-based data. Despite the ongoing advancements in this field, it remains challenging to recognize complex human activities for a specific domain. To bring attention to this issue, the "Third Nurse Care Activity Recognition Challenge" gathered accelerometer data from smartphones to estimate daily nurse care activities. The main challenge was handling the noisy and inconsistent dataset, which is a persistent issue in any real-life data. Also, each activity depends on both the subject and the receiver, making its recognition more complex. Our team, Team Alkaline, used high pass and low pass filters to reduce noise, adopting an overlapping windowing technique. Then we extracted features in multiple domains to derive the necessary information required for classifying the human activities more accurately. Later on, we used a feature selection method to select the most significant features. We applied Random Forest (RF) classifier for training and achieved 99.0% accuracy on the validation set.

CCS CONCEPTS

• **Computing methodologies → Machine learning**; **Machine learning approaches**; *Feature selection.*

KEYWORDS

Human Activity Recognition; Nurse Care Activity Recognition; Feature Extraction; Random Forest

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1 INTRODUCTION

There has been a remarkable improvement of microelectronics and processing units throughout the past decade, enabling sensors and mobile devices with unbeaten characteristics. Their high computational power, small size, and minimum cost allow people to interact with the devices as part of their daily living, which has facilitated the process of automatic identification of Human Activities. At present, researchers are exploiting different sensors data for solving the challenges of Human Activity Recognition (HAR). Among them, Accelerometers are extensively used to estimate idle time, physical activity, physical activity energy expenditure, and sleep-related behaviors [8].

The recognition of human activities has brought significant advantages in different areas, especially in healthcare. Comparatively recognizing simple activities like walking, sitting, etc., is more straightforward than recognizing nurse care activities since it mostly involves complex activities like bathing, exercising, etc. Also, a single activity includes both nurse and the patient to be performed. Depending on the patient condition, same activities are performed in different ways. Recognizing nurse care activities make the caregiver's work more efficient and focus their efforts on the core activities instead of other activities such as documentation. By identifying the activity a nurse is occupied in, we can partially automate the documentation process to reduce time spent on the task. Monitoring the care plans ensure that all care activities have been conducted for each patient [9].

Sozolab has organized a competition for the third time, 'Third Nurse Care Activity Recognition Challenge,' to achieve this goal. The organizers provided raw reading based on the accelerometer data collected from the smartphone to recognize the daily activities of the caregiver/nurse in a healthcare facility more accurately [4].

Working with the given dataset, there are numerous difficulties since it requires a considerably huge amount of cleaning for being highly noisy, imbalanced, and inconsistent data. On another note, the data was collected using mobile devices, the position and the orientation of these devices while sampling was arbitrary. Therefore, there is a possibility of axes being interchanged among themselves, affecting the extracted features. Moreover, many of the activities are not clear, and they might indicate a range of activities. Multiple activities were so similar to each other that it was hard to differentiate. In this paper, we, Team Alkaline, proposed an algorithmic pipeline including different preprocessing and feature extraction techniques for better classification of the complex nurse care activities.

We describe the paper as follows: Section 2 covers the Related Works. Section 3 explains the Methodology of our approaches, which includes some subsections. Section 4 includes the explanation and figures of experimental results, and with that, we conclude the paper in Section 5.

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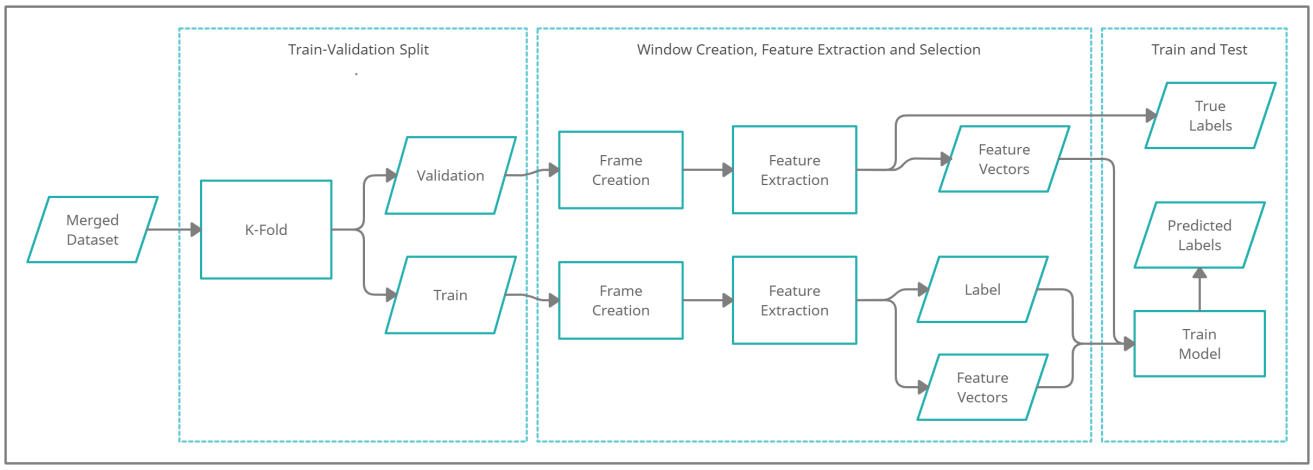


Figure 1: Train-Test Pipeline

2 RELATED WORK

Acknowledging the impact of activity recognition, researchers are analyzing different methodologies and approaches to find a better way to recognize the human activities accurately. Previous works on Nurse Care Activity Recognition Challenges have proposed different approaches on preprocessing and feature extraction depending on activities and the characteristics of the sensor data.

In [7], J.R.Kwapisz *et al.* mentioned that they divided the data into 10-sec segments after that generated features, based on the 200 readings encompassed within each 10-sec segment since one cannot use standard classification algorithms directly on the raw time-series accelerometer data.

In [6], the authors showed that using a simple approach for K-Nearest Neighbour (KNN) along with necessary feature extraction achieved better results than deep learning algorithms. In another work [5], M.S Irbaz *et al.* also emphasized working on different traditional machine learning (ML) algorithms and measuring the performance on traditional machine learning algorithms with the integration of frequency-domain features. They used overlapping sliding windows with hop-size 10, and KNN gave the best result for their approach. They concluded that machine learning classifiers with the collaboration of proper feature extraction might be helpful to identify the activity with a low computational cost. P.Basak et al.[3] processed the data by filtering noise, applying windowing technique on time and frequency domain to extract various features from lab and field data distinctly. After merging the data, they applied a unique cross-validation technique to find out the best-performed model using Random Forest (RF). However, H.Matsuyama et al.[10] used a different approach. They mainly focused on not letting the model overfit. They also achieved the best result using RF.

In [1], the authors summarized the work of the participants of the Second Nurse Care Activity Recognition Challenge and specified that ML algorithms worked way better than deep learning algorithms. Their observation was that RF classifies the activity better than other ML algorithms due to its ability to handle large datasets with missing, noisy values.

3 METHODOLOGY

In this section, we briefly discuss our algorithmic pipeline used to solve the challenge. We divide the proposed approach into four major sub-parts, which are described in detail in the following subsections:

3.1 Preprocessing

The dataset available for this challenge was divided into two parts. One consisted of the accelerometer samples for 12 different users, and another consisted of the labeled time windows.

At first, we dropped the samples consisting of missing values and duplicates. We formatted the timestamps of accelerometer samples and labeled windows to a standard timezone before merging. We also swapped the start and finish times of the rows with finish time greater than start time and added 59 seconds with the start time for the rows with equal start and finish time in the labeled windows before merging. This step helped us to gain a significant amount of increase in labeled accelerometer samples.

The positional data for the x, y, and z coordinates were scattered and noisy in the spatial domain. To process the raw data, we used normalized high pass and low pass filters. High pass filter passes the signals with a frequency higher than the cutoff frequency and attenuates the lower frequencies, whereas Low pass filter does the opposite. This filter passes the signals lower than the cutoff frequencies and attenuates the higher frequencies. We applied these filters in three different setups for analyzing their impact in classification: (1) Using only a high pass filter; (2) Using only a low pass filter; (3) Using a high pass filter followed by a low pass filter; (4) Using a low pass filter followed by a low pass filter.

3.2 Frame creation

We applied an overlapping windowing technique to create the frames from the readings of the raw accelerometer. We created two separate feature vectors, using 8-second and 10-second frames to get samples for at most a single instance of a single activity type per frame. We used a hop size of 10 data objects and subsequently

Table 1: The set of features extracted from the accelerometer sensor data. Here, X and Y is used to indicate data sequence of an axis in a single frame, N is the length of the data sequence in a frame, x_i and y_i indicates i^{th} sample of X and Y, $\bar{x}$ and $\bar{y}$ indicates mean of X and Y respectively and σ indicates variance. In the formula of mode obs. is the short form for observation.

Features	Formulas
Maximum	X_{max}
Minimum	X_{min}
Mean	$\frac{1}{N}\sum_{i=1}^{N} x_i$
Median	x such that $F(x) = 0.5$
Mode	If N is even then $\tilde{x} = \frac{(\frac{N}{2})^{\text{th}}\text{ obs.}+(\frac{N+1}{2})^{\text{th}}\text{ obs.}}{2}$ If N is odd then $\tilde{x} = \frac{N+1}{2}^{\text{th}}$ obs.
Standard Deviation	$\sqrt{\frac{1}{N}\sum_{i=1}^{N}(x_i - \bar{x})^2}$
Variance	$\sum_{i=1}^{n}(x_i - \bar{x})^2$
Skewness	$\frac{1}{N}\sum_{i=1}^{N}\frac{(x_i-\overline{x})^3}{\sigma^3}$
Kurtosis	$\frac{1}{N}\sum_{i=1}^{N}\frac{(x_i-\overline{x})^4}{\sigma^4}$
Signal Power	$\frac{1}{N}\sum_{i=1}^{n} x_i^2$
Root Mean Square	$\sqrt{\frac{1}{N}\sum_{i=1}^{N}(x-\overline{x})^2}$
Correlation	$\frac{\sum_{i=1}^{N}(x_i-\overline{x})(y_i-\overline{y})}{\sqrt{\sum_{i=1}^{N}(x_i-\overline{x})^2(y_i-\overline{y})^2}}$

trained two different classifiers and observed their performances. From the observation, using 8-second frames yielded better results. So, for our model, we selected 8 seconds as our window size for creating frames from the accelerometer samples.

3.3 Feature Extraction and Selection

After creating the time frames, we extracted different features for each frame to train our model. We initially extracted 36 features (12 for each axis) shown in Table-1. Among these different types of features, statistical features occupied a major space, since according to previous studies, statistical features work effectively for human activity recognition [2]. There are basic features like maximum, minimum to understand the waveform; mean, median, and mode to measure the central tendency; standard deviation and variance to measure the variability of the data; skewness, and kurtosis to characterize the data distribution. We also extracted correlation among the axes to measure the relation between axes, signal power, and the root mean square for each axis. Later on, we ranked the features according to the information gain and selected the top 29 features for our model among the 36 features we extracted initially. The features we dropped were correlation among every pair of axis, variance and standard deviation for y-axis, kurtosis, and skewness for x-axis. Selected features were extracted from each frame and passed for training and testing (Figure-1). The activities of the challenge were too generalized to detect specific features to extract from a specific activity.

3.4 Classification

We explored three different Machine Learning algorithms: RF, K Nearest Neighbour (KNN), and XGBoost. RF is an ensemble algorithm that relies on the results of multiple classifiers of the same or different kinds, works better for missing values, high dimensions, and the high number of classes. XGBoost is also an ensemble algorithm that uses many decision trees as base learners. On the other hand, KNN does not need any training data points for model generation and has no assumption for underlying data distribution.

4 EXPERIMENTAL RESULTS

4.1 Dataset Description and Implementation Detail

The dataset[4] for this challenge was collected using smartphones carried by nurses while conducting their daily works at healthcare facilities. The position and orientation of these smartphones were arbitrary. It consisted of accelerometer readings for 28 different activities, which were divided into four groups as shown in Table-2. The size of the dataset got reduced from 27,448 rows to 26,815 rows after dropping the rows with missing values, and no samples for activity type 21 were available in the labeled dataset.

After merging accelerometer samples with the labeled dataset, there are 2,074,510 samples. As shown in Table-2, there is a noticeable amount of imbalance in every class. Moreover, for activity types 21, 23, and 27, no labeled samples were found after merging. We can not extract any feature for the activity types 20, 21, 23, 25, 26, and 27 due to the very small number of samples of those activities were present.

For splitting the dataset into train set and validation set, we applied stratified 5-fold cross-validation to ensure every class's presence with an equal percentage in every fold. To evaluate the performance of the proposed approach, we use four metrics, namely Accuracy, Precision, Recall, and F-score in this paper. While filtering the data, the cutoff frequency used for the low pass filter was 10hz, and for high pass filter, it was 50hz.

4.2 Result Analysis and Discussion

In this section, we analyze the performance of our proposed approach in recognizing complex nurse care activities. In order to remove the noisy data, we filter the dataset in different ways. The impact of applying different filtering mechanisms is shown Figure-2. In Figure-2a, we observe the accelerometer readings for x, y, and z axes without any filters. Applying normalized low pass filter, we achieved smoother data reducing the effect of sudden changes as shown in Figure-2d. On the other hand, a normalized high pass filtered data gave a clear idea of the area of changes as it attenuated areas of sudden changes in acceleration as shown in Figure-2e. We compared two filters, one with the data filtered using normalized high pass filter and another using data filtered with normalized low pass filter. We also analyzed the data applying normalized low

Table 2: Activity types divided into 4 broader groups and their corresponding IDs (on the left) according to the challenge dataset. Total number of samples for each activity in the merged dataset is given within the parenthesis.

Direct care	Residence cleaning	Documentation/Communication	Others
1. Vital (8999) 2. Meal/medication(392555) 3. Oral care (2041) 4. Excretion (245030) 5. Bathing/wiping (3403) 6. Treatment (2412) 7. Morning gathering/ exercises (1052153) 8. Rehabilitation / recreation (1342) 9. Morning care (23041) 10. Daytime user response (45765) 11. Night care (11430) 12. Nighttime user response (62505) 13. Family/guest response (34097) 14. Outing response (157432) 19. Get up assistance (20774) 20. Change dressing assistance (40) 21. Washing assistance (0) 27. Emergency response such as accident (0)	15. Linen exchange (1199) 16. Cleaning (2944) 23. Preparation and checking of goods (0) 24. Organization of medications (948)	17. Handwriting recording (1619) 18. Delegating/meeting (1483) 22. Doctor visit correspondence (1848) 25. Family/doctor contact (6)	26. Break (83) 28. Special remarks/ notes (1361)

pass filter followed by normalized high pass filter. It is evident from (Figure-2c) that the combination of both filters work better. We got a better understanding of the area of changes from a smoother trend. This is because the normalized low pass filter smoothened the data, and then the normalized high pass filter attenuated the area of changes. Therefore, we choose to filter the data by applying normalized low pass filter followed by a normalized high pass filter before extracting the features.

Table 3: Performance comparison of the proposed algorithm in different classifiers without any filtering

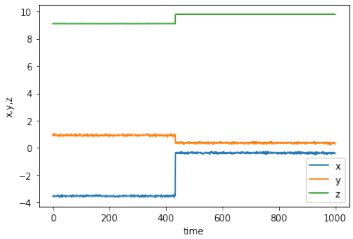
Classifier	Accuracy	Precision	Recall	F1-Score
Random Forest	89.43	88.65	89.43	88.94
XGBoost	87.5	86.88	87.496	86.55
KNN	63.63	59.81	63.33	61.19

Table 4: Performance of comparison proposed algorithm in different classifiers using highpass and lowpass filters

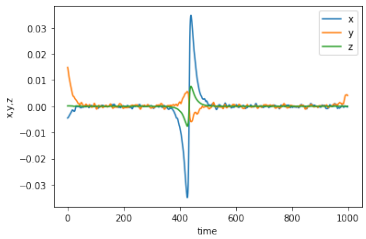
Classifier	Accuracy	Precision	Recall	F1-Score
Random Forest	99.00	99.06	99.02	99.02
XGBoost	97.05	97.07	97.05	96.99
KNN	48.25	45.78	48.25	46.57

Moreover, to demonstrate the effect of filtering in classification, we compare the performance of classifiers in Table-3 and Table-4. We generate the results of different evaluation metrics using RF, KNN, and XGBoost with unfiltered data and filtered data to get an idea of which model performs better in this case. As shown in Table-3, without any filtering, XGBoost and RF classifiers performed well compared to KNN. After applying the filter in the data, we obtained significant improvement in results except for KNN classifier shown in (Table-4). KNN performed comparatively worse on the filtered data because filtering has modified the data in the spatial domain (Figure-2), and KNN works with the relative data points in the spatial domain. In terms of both unfiltered and filtered data, RF performed slightly better than XGBoost due to its robustness to the outliers [11]. Therefore, we further analyze our result only using RF as a classifier.

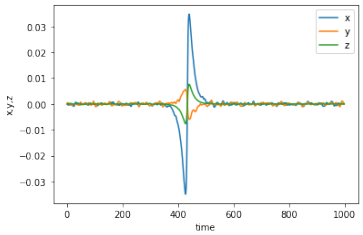
Our proposed approach of recognizing complex nurse activity includes data filtering, creating appropriate windows, and extracting and selecting the important features from the raw data. We obtained the 99% accuracy using RF. The reason behind such high accuracy is that we are able to remove the noise from the dataset properly and also extract the important features required for the discrimination among the different activities. Although the dataset was imbalanced, the value of precision, recall, F-measure demonstrates that our proposed approach can also identify the minority classes to some extent. For better understanding the class-wise performance, we show the confusion matrix for our final model in Figure-3. We observed that in the broad activity category: Other Activities, activity 28 gained an accuracy of 50%. We got an accuracy of more than 99% for the broad activity categories: Activities of Residence Cleaning and Documentation/Communication Activities. On the other hand, in the broad category: Activities of Direct Care, we noticed a comparatively higher amount of false positives for activities 2, 10, 12, and 13. The reason behind this could be that



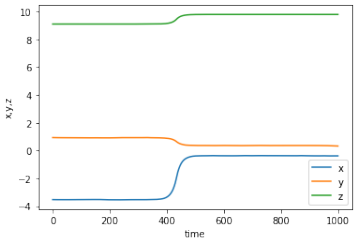
(a) data before using filter



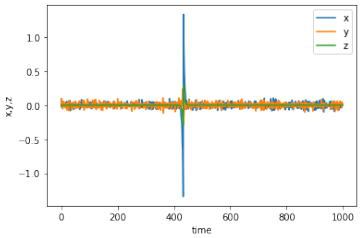
(b) data after using high pass followed by a low pass filter



(c) data after using low pass followed by a high pass filter



(d) data after using a low pass filter



(e) data after using a high pass filter

Figure 2: Data before and after using filters

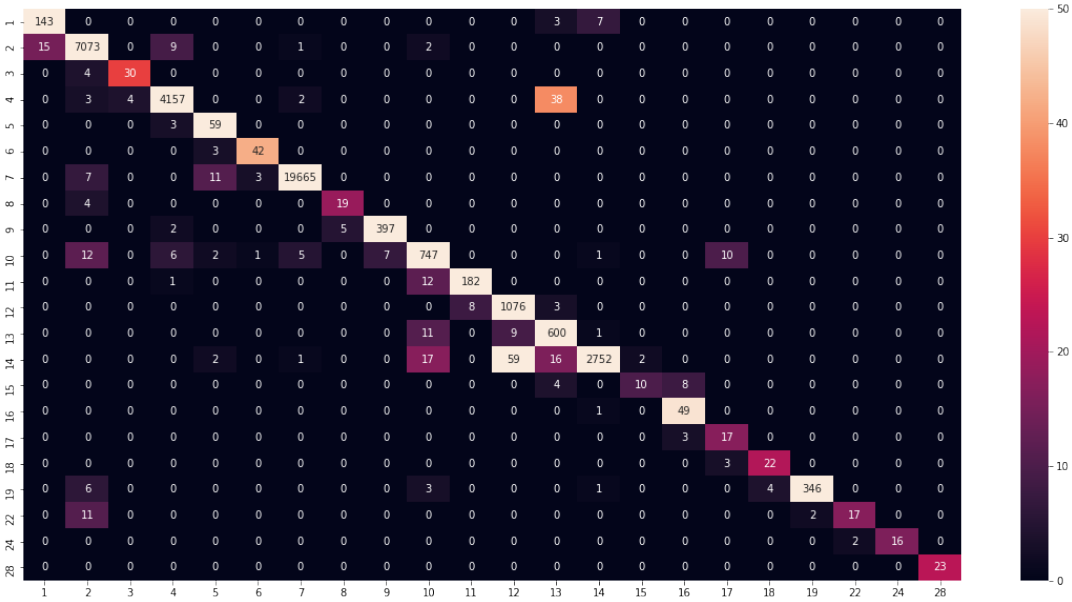


Figure 3: Confusion Matrix for the validation set using Random Forest Classifier

the activities were so similar, the classifiers could not differentiate them properly.

5 CONCLUSION

In this paper, we proposed a method for recognizing the complex nurse care activities based on accelerometer data. At first, we preprocessed the data for removing noisy instances and then extracted and selected the features in multiple domains from each frame. Among the classifiers used in our experimentation, the proposed method obtained the highest performance using Random Forest classifier. Although using the filters in processing the dataset played a good role in increasing classification performance, finding a more convenient figure for the cut-off frequency through testing should be investigated later. In future, we plan to explore more appropriate frame sizes according to the activities, improve the feature extraction and feature selection process. The recognition result for the testing dataset will be presented in the summary paper of the challenge [b].

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6 APPENDIX

Table 5: Summary of resources and other information

Used sensor modalities	**Accelerometer**
Feature used	Max, Min, Mean, Median, Mode, Variance, Standard Deviation, Kurtosis, Skewness, Signal Power and Root Mean Square for each axis (x, y, z)
Programming language and libraries used	Python 3, Math, Numpy, Pandas, Scipy, Sklearn, Pickle, Seaborn, Matplotlib
Window size and Post Processing	Frame size: 8 seconds Minimun data objects in a window: 10 Hop size: 10
Training and testing time	Training time: 1136.29s Testing time: 8.85s
Machine specification	Google Colab